

From Firefighting to Zero-Touch:

How AIOps Empowers SREs to Build Resilient, Business-Driven Platforms

June, 2025

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Speakers



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Agenda

From Firefighting to Zero-Touch: How AIOps Empowers SREs to Build Resilient, Business-Driven Platforms

1. Journey to AIOps: A Comic
2. Demo: Zero-Touch AIOps
3. Conclusion & Takeaways

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Journey to AIOps: A Comic

Key Challenges of Modern IT

IT modernization creates explosive growth in the scale and diversity of the IT estate. This deluge of complexity is inhibiting organizations from operating efficiently and realizing the full value of their initiatives.



Increasing IT refreshes

Infrastructure refreshes, app modernization, and cloud migrations continuously evolving the IT landscape and requiring modern tooling to monitor and contextualize data



Dynamic Architectures

Growing number of hybrid, multi-cloud and microservices that can't be quickly monitored, diagnosed and troubleshooted for issues



Service Availability

Maintaining high levels of service availability with the ability to detect service impacting issues proactively



Intelligent DevOps

Automating the path from development to production with focus on shift-left analytics and monitoring



Data Deluge

Increasing data volume, velocity and variety, overwhelming capacity and manual processes



Information Silos

Teams constantly needing to switch across several tools to make sense of critical operational data

Market Backdrop

43% of IT decision makers fear their IT infrastructure won't be able to handle future data demands[1]

90% of mean time to repair (MTTR) is the time taken to identify the source of the problem[2]

96% of orgs run some cloud-native apps across multiple environments, increasing the complexity of their architecture[2]

Sources: 1: Dell Technologies, 2: ZK Research

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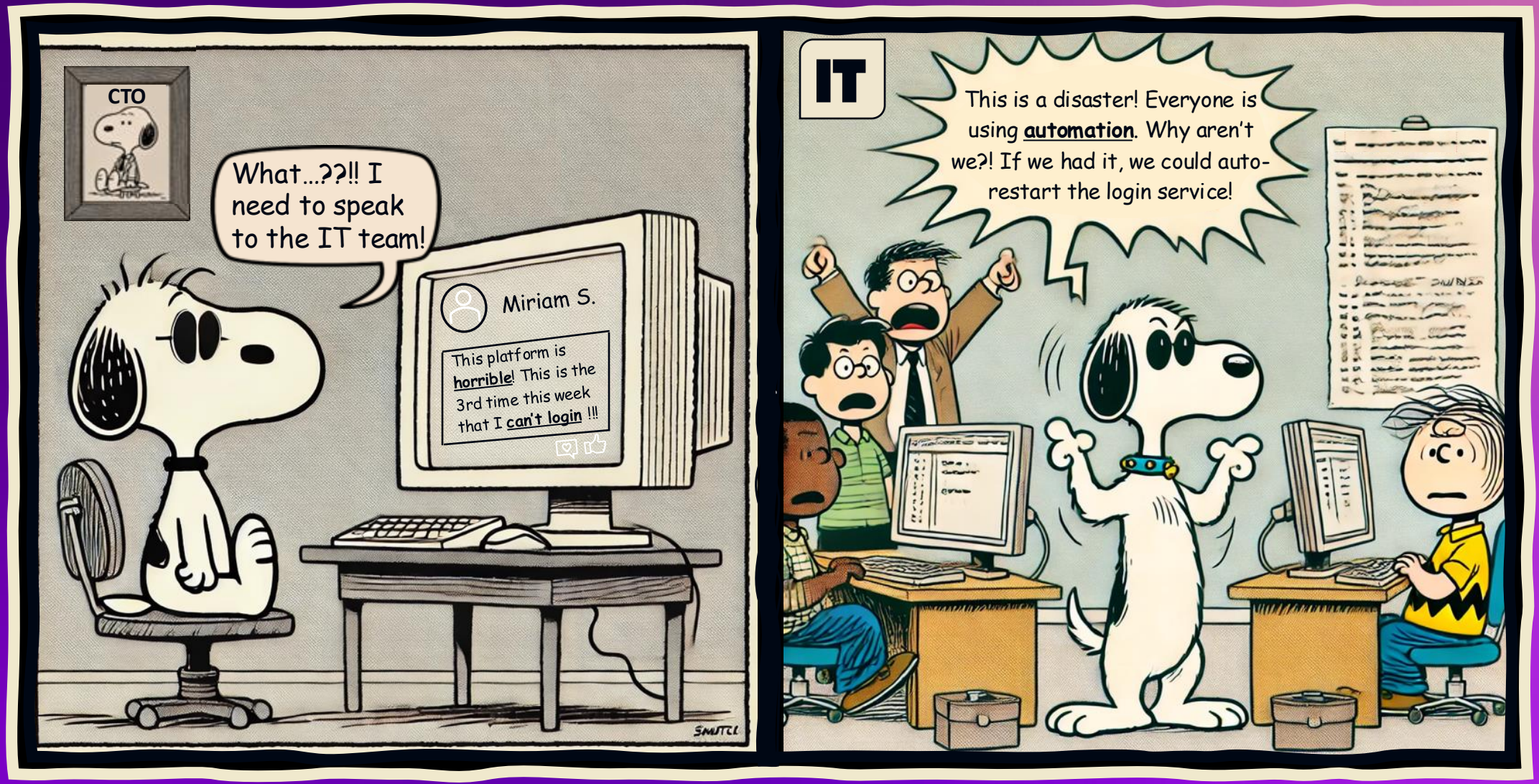
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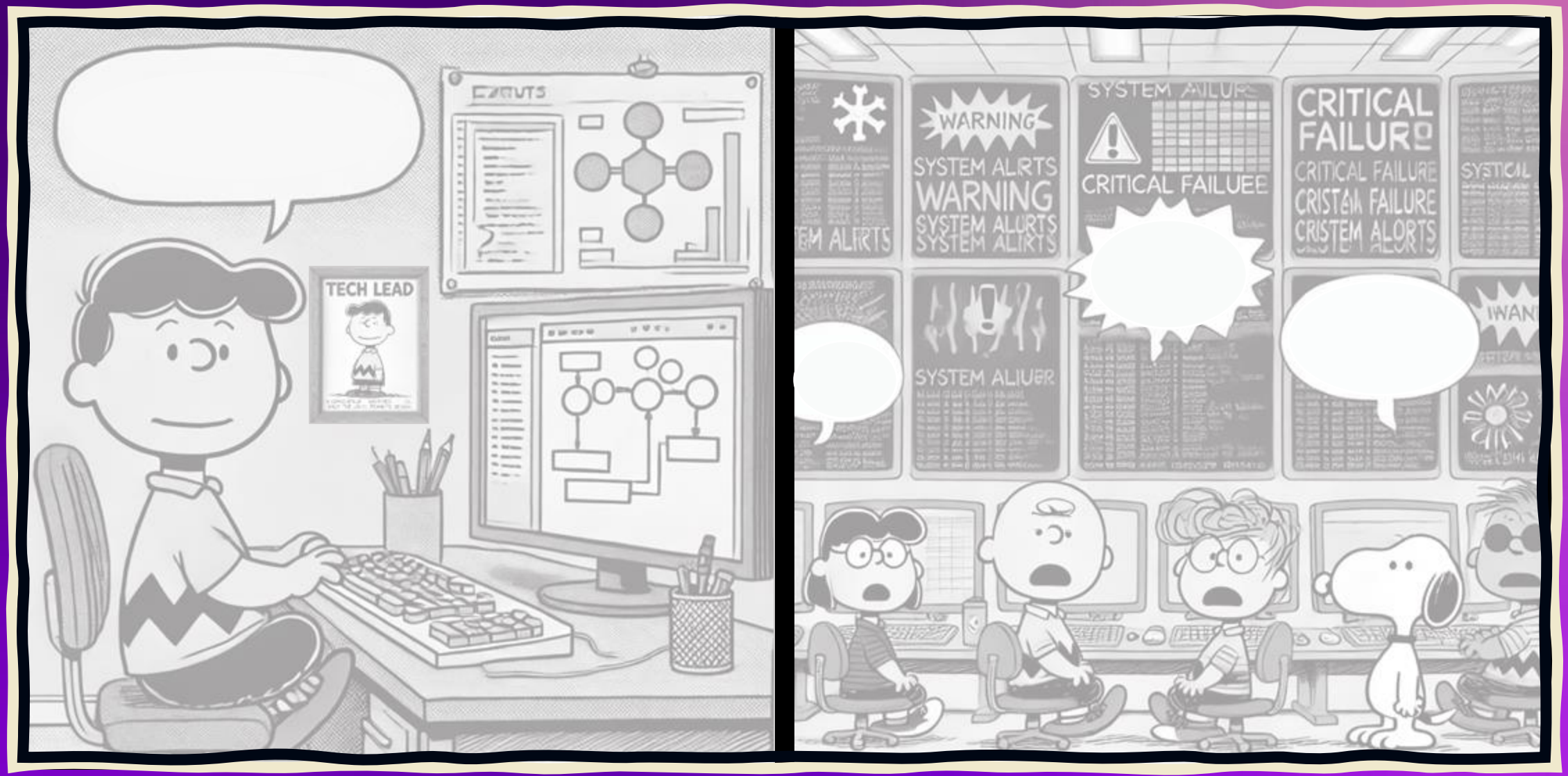
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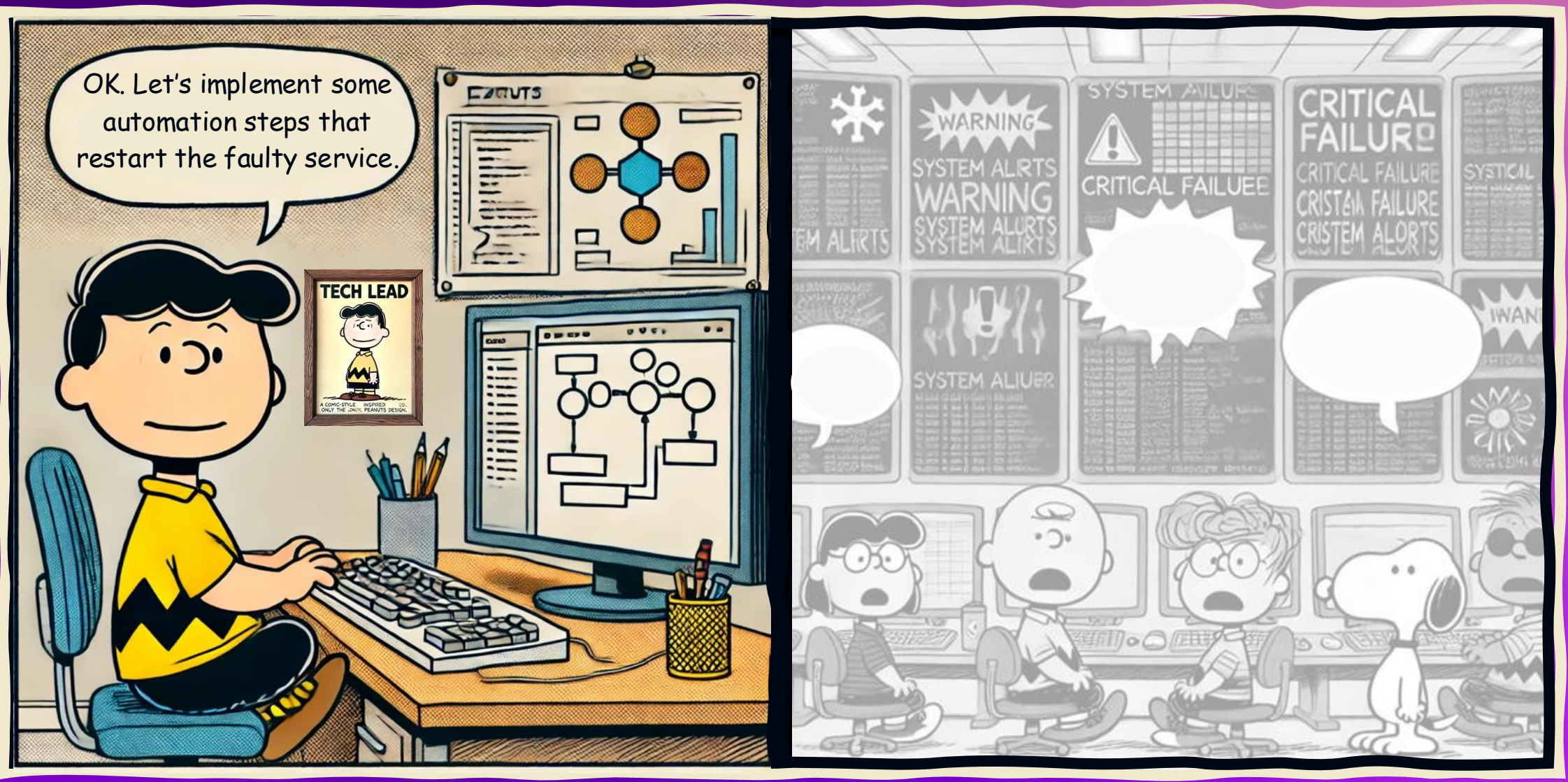
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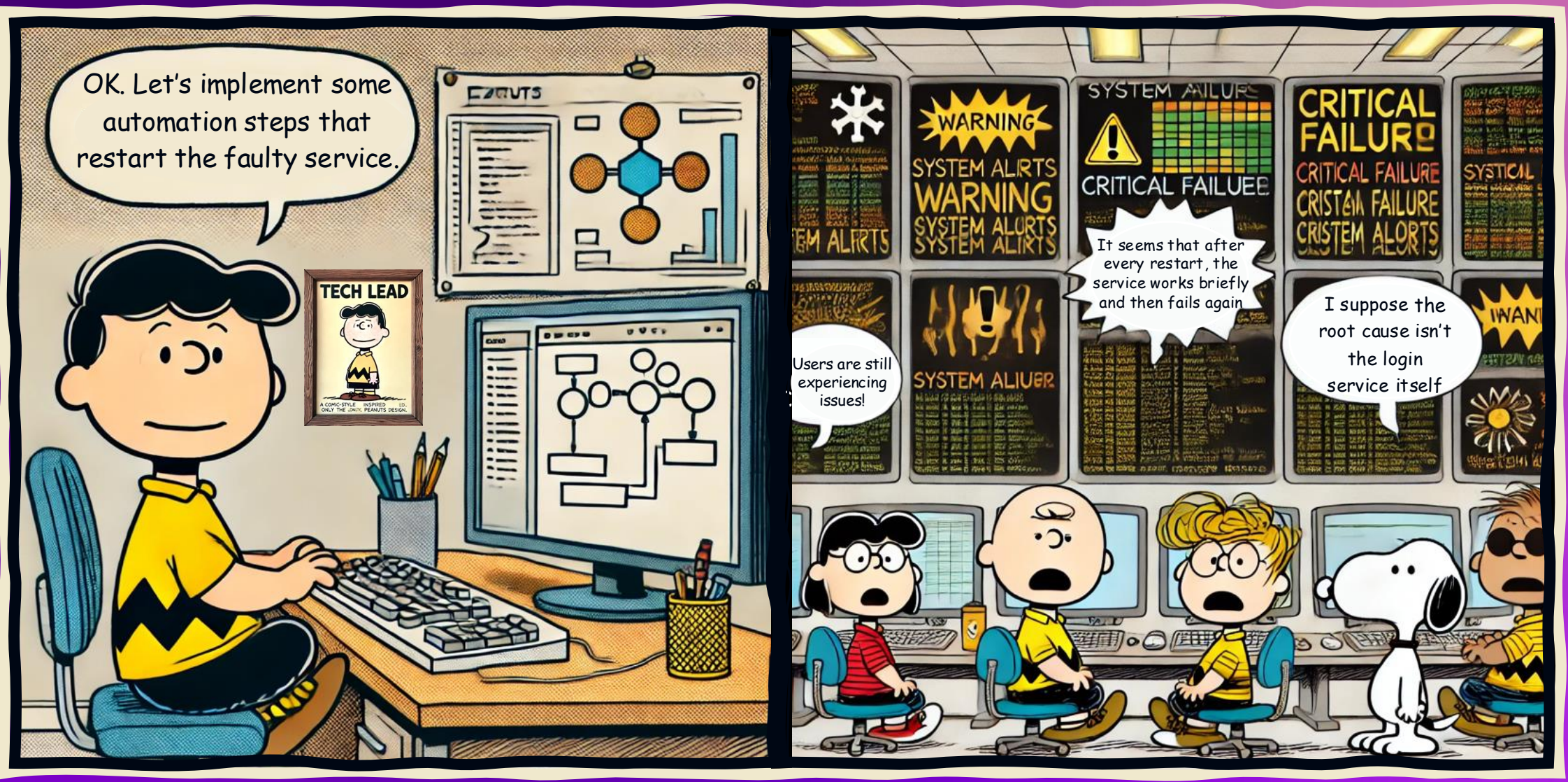
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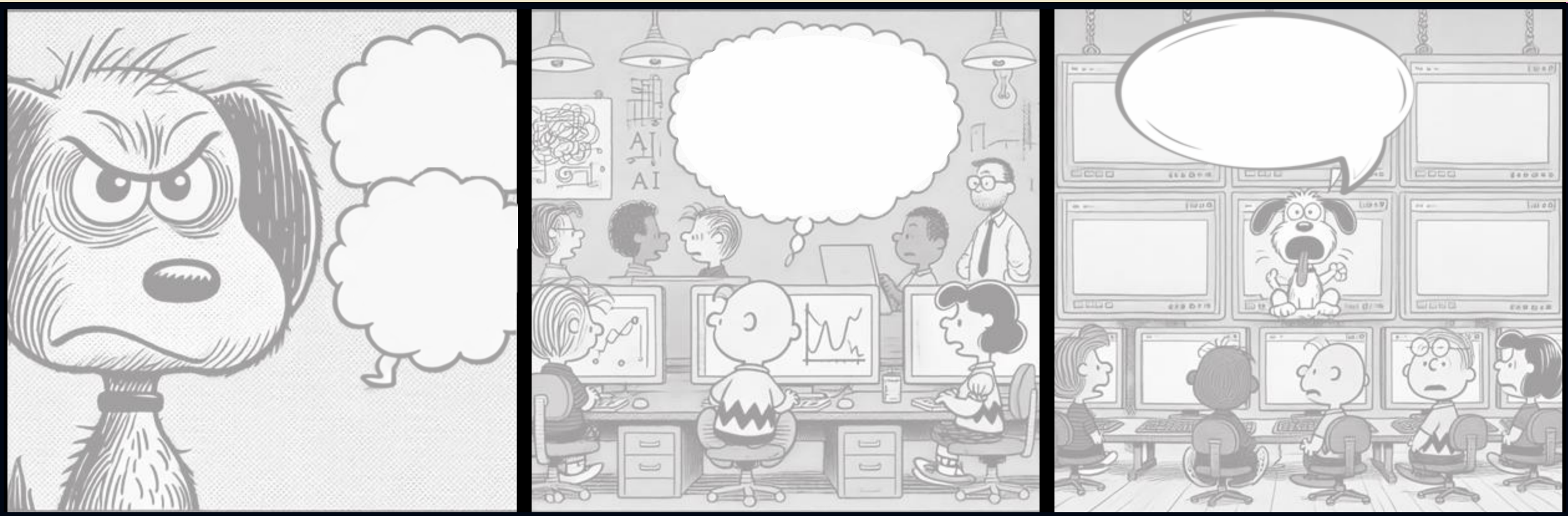
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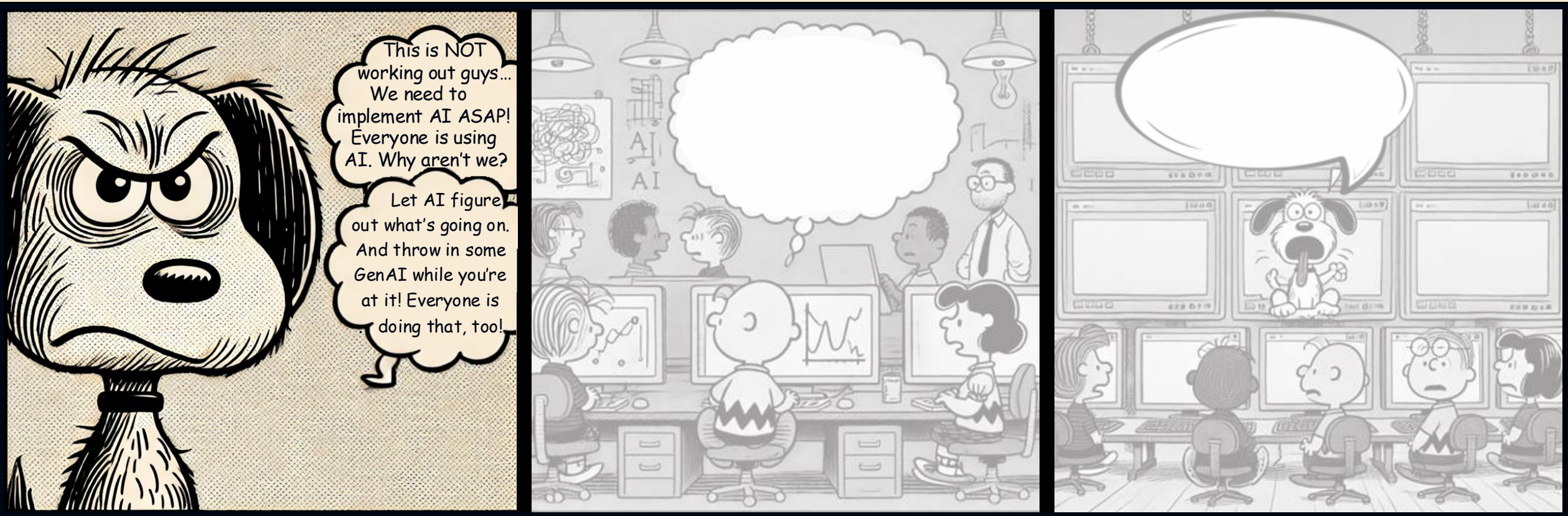
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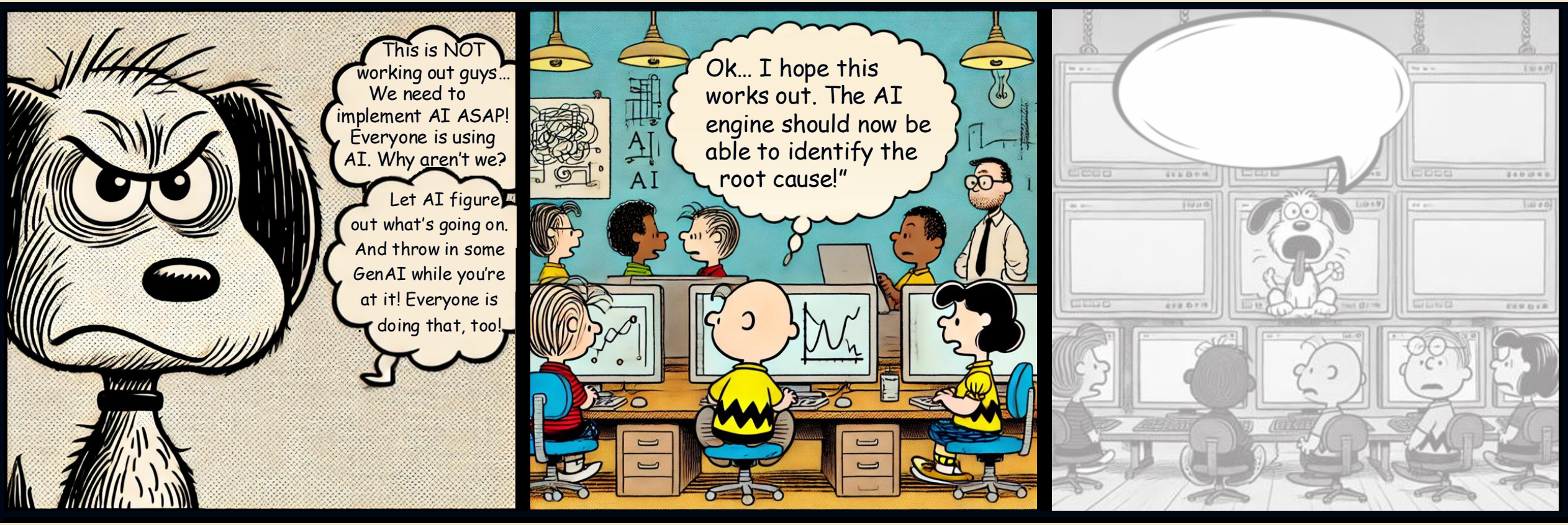
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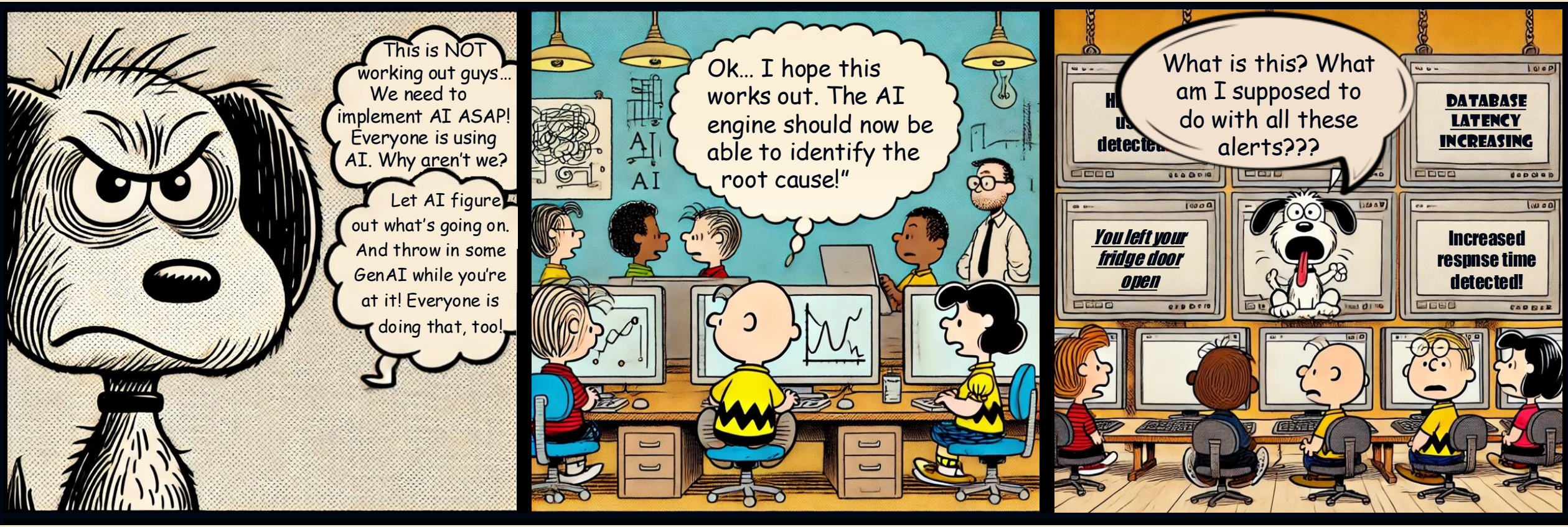
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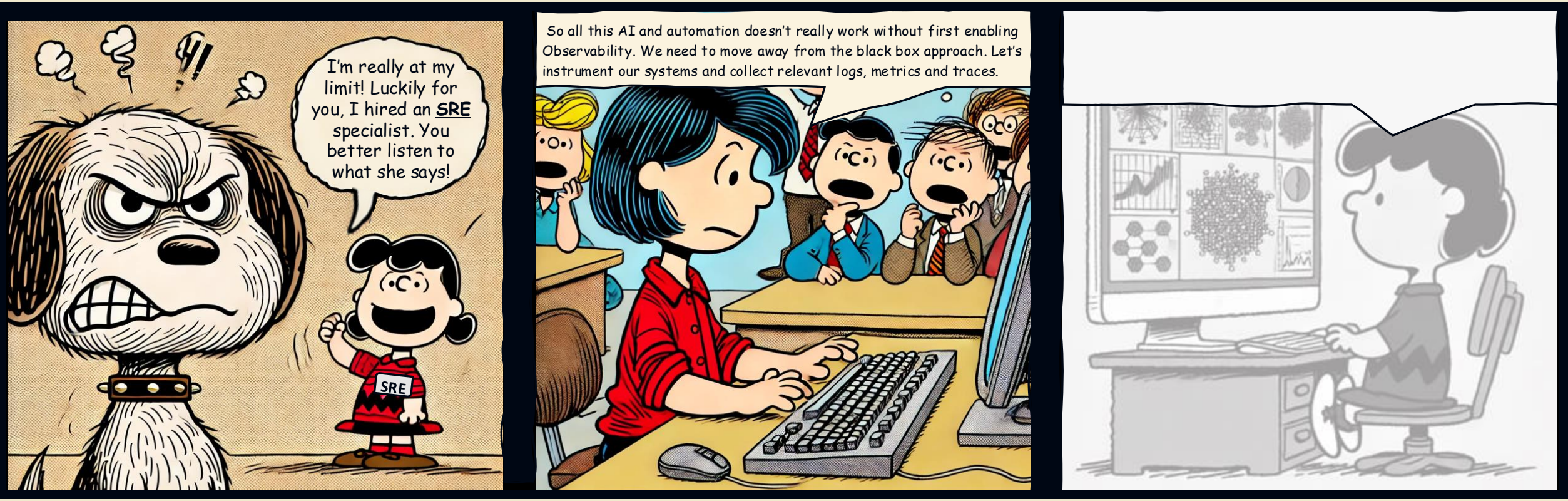
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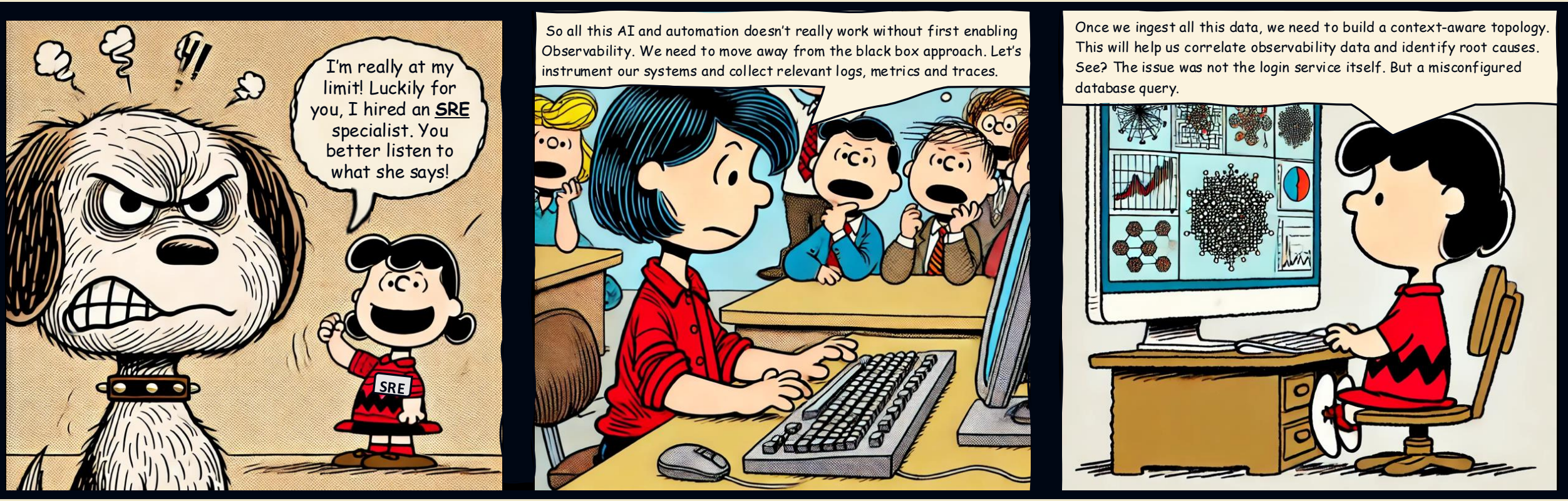
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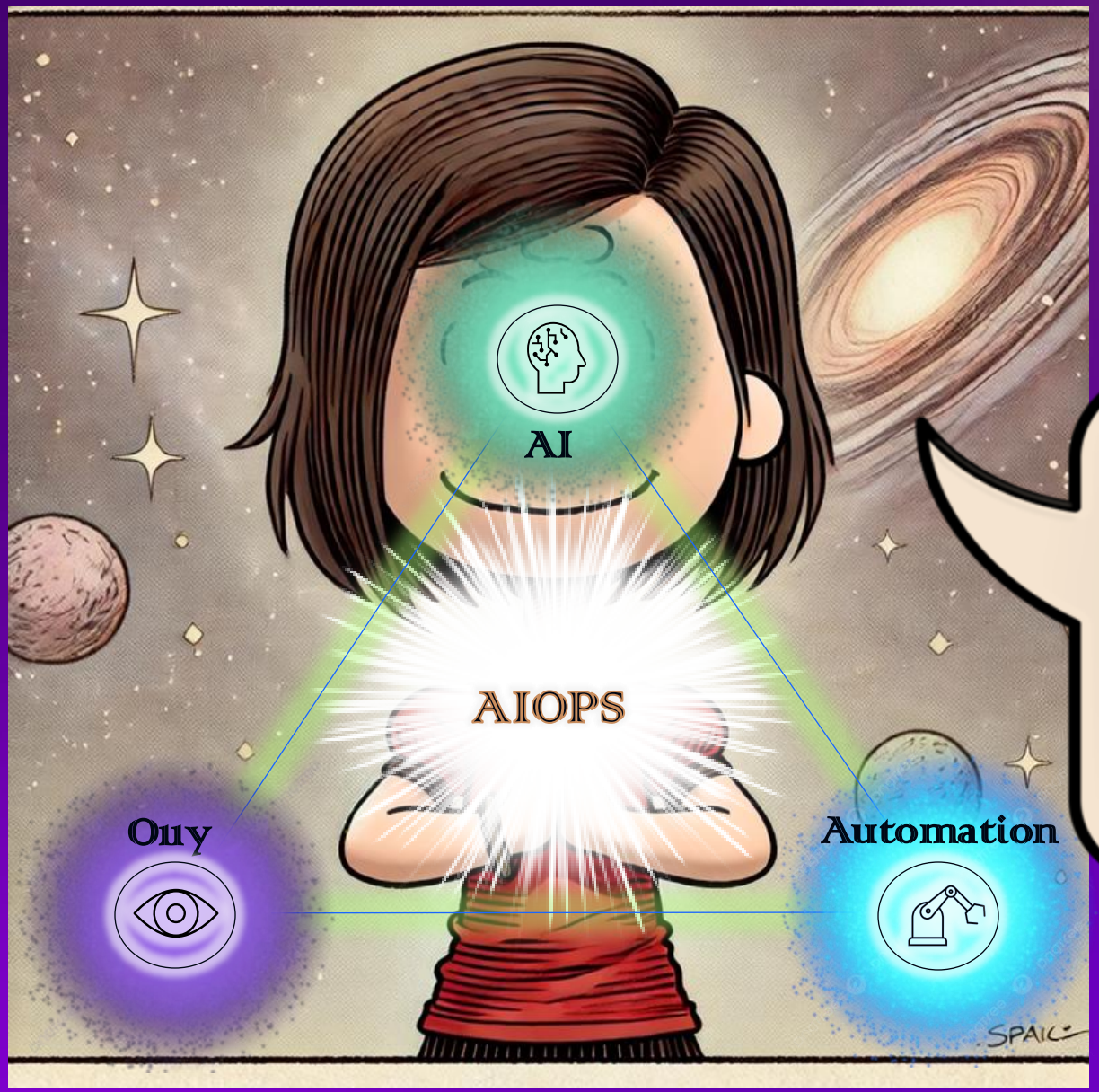
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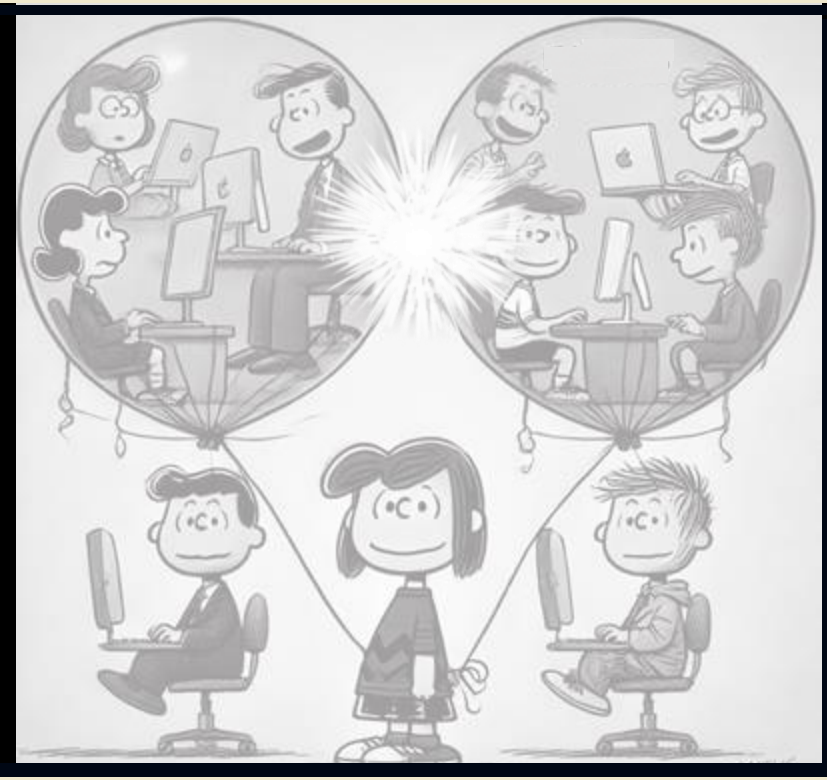


Now that we know what the root cause is, and now that we have the historical **observability** data, we can use **AI** to predict this failure before it actually occurs! Only then can we apply your **automation** script that triggers the self-healing process. And this is how you build a full-fledged **AIOPS** solution!

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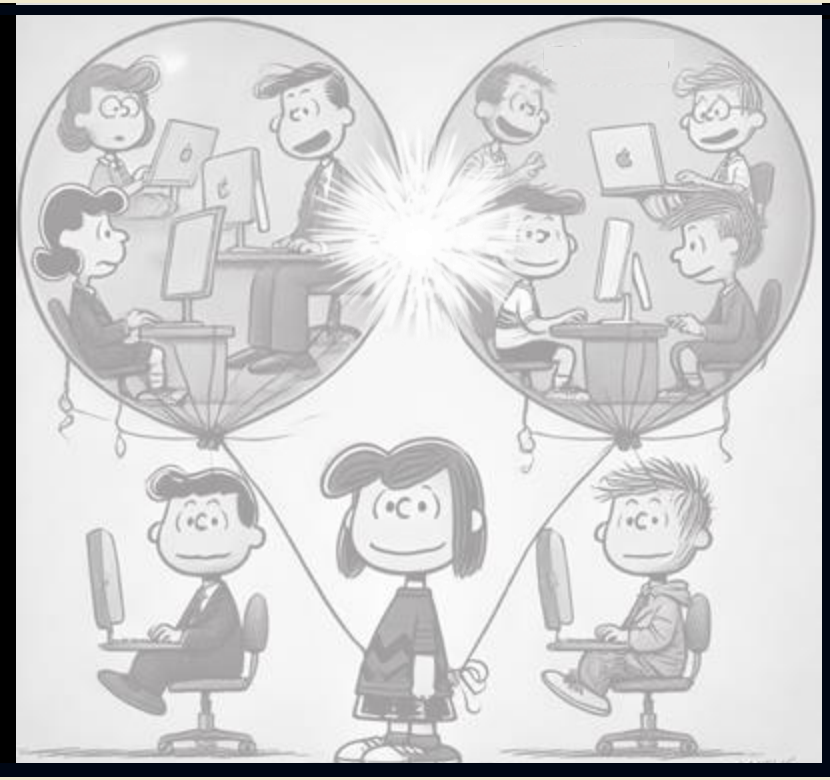


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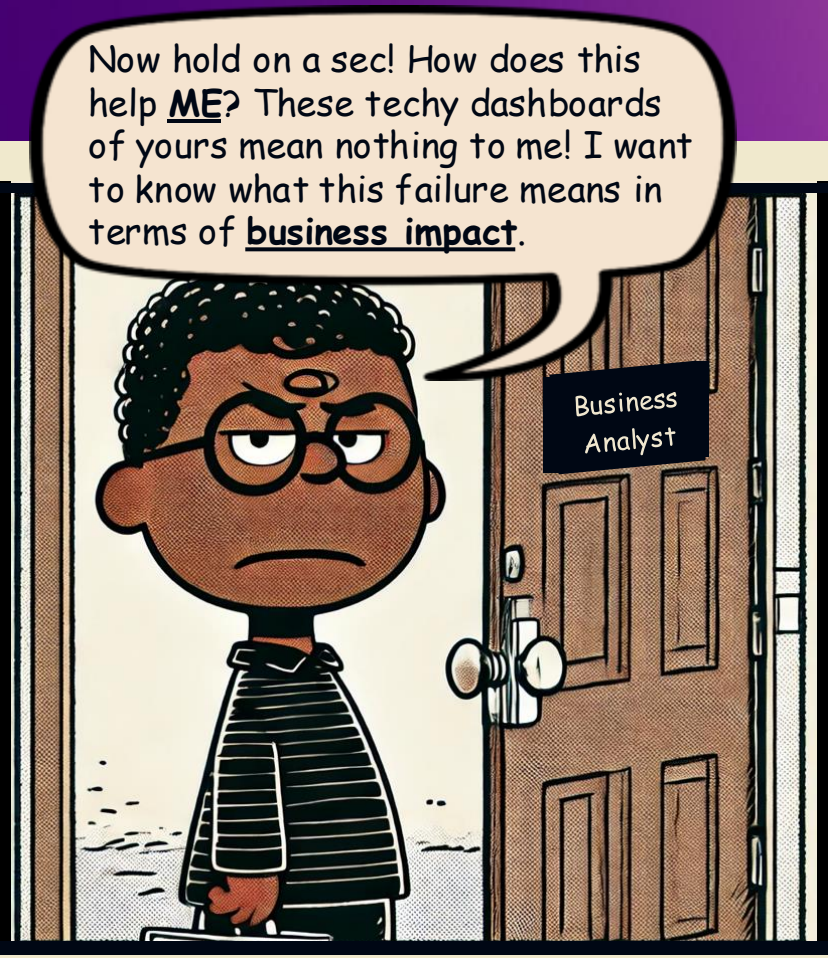


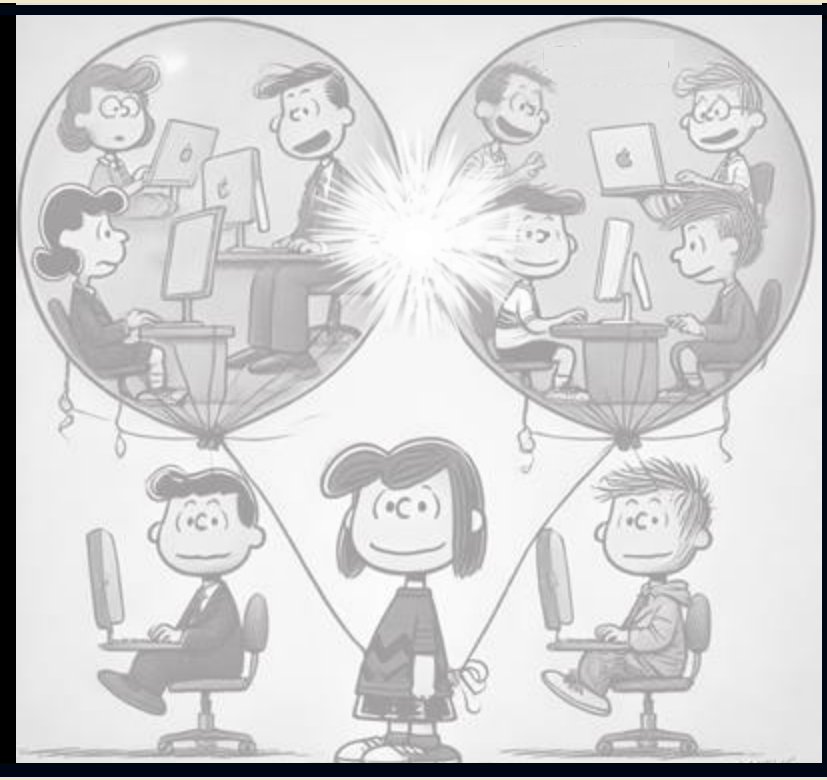


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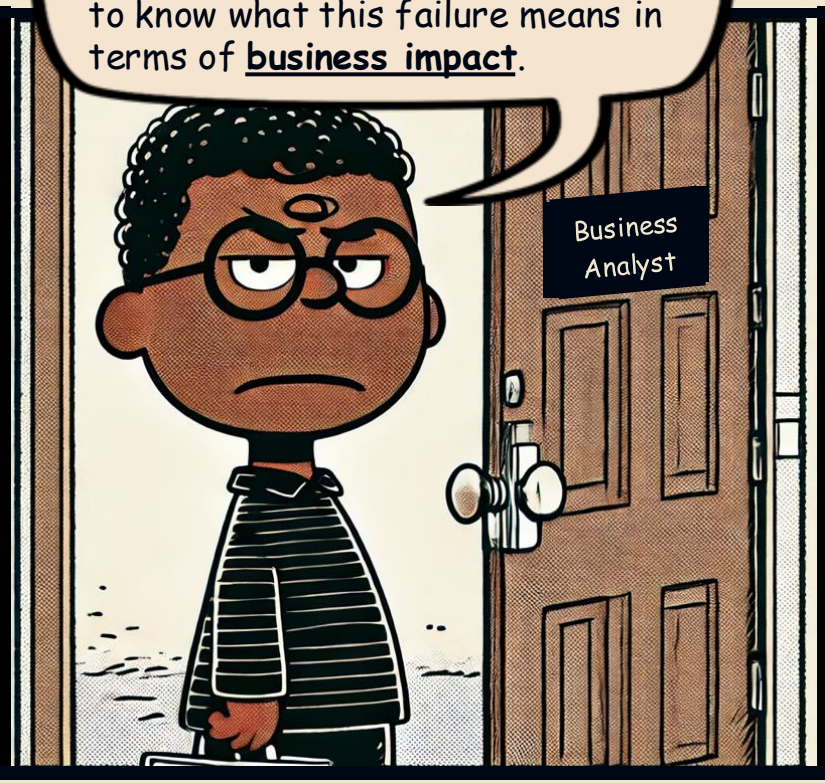




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AIOps & Observability Adoption Journey

From Monitoring to Zero-Touch Operations

Observability Transformation Journey

Reactive
Monitoring

Monitor the components and make sure that they are working. Check performance, capacity, availability, events, alerts (siloed visibility).

Aware
Observability

Determine **why** the failure occurred via the golden triangle of o11y (metrics, traces, logs). Start aligning insights from observability with Business perspective.

Proactive
Full-stack / Contextual Observability

Contextualize telemetry data, root cause analysis, **topology-based correlation**. **Analyze trend** and **act** faster, with ability to understand and assess customer journey health.

Preventive
Intelligent / Predictive Observability

Leverages **machine learning** and advanced analytics to **predict** potential issues and trends based on historical data. Aid with capacity planning, green coding, and cost/experience optimization.

Automated
Federated / Autonomous Observability and AIOps

Zero-touch operations with self-healing. Any data available to any consumer on demand. Highest **business impact** at lowest possible cost.

Increasing Business Insights by Strengthening the Correlation between IT and Business Metrics

Increasing IT Insights

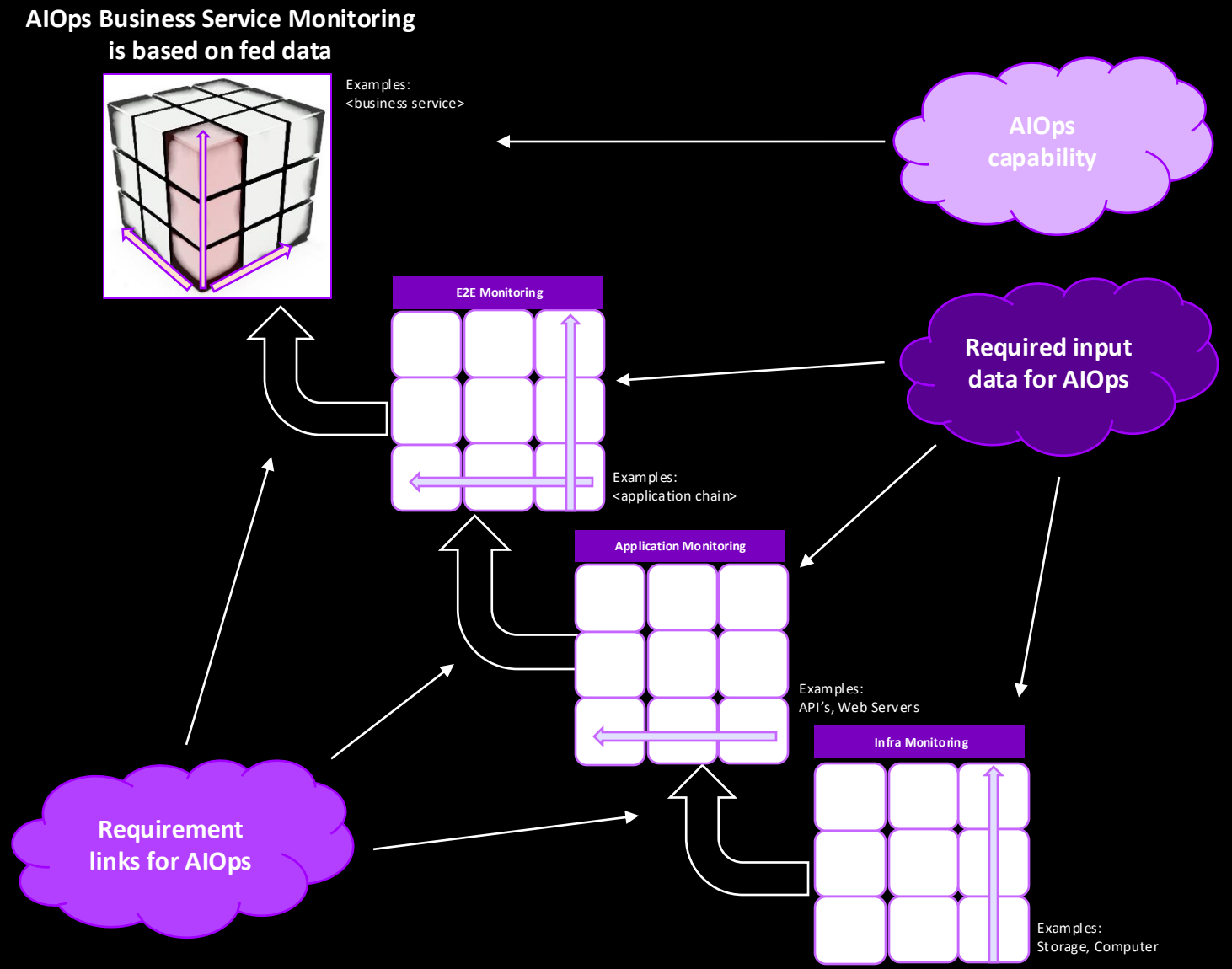
Expected Outcomes

- **Leverage AI and automation** to drive efficiency and agility
- **Streamline resource allocation**
- **Enhance Availability, Performance and Resiliency** across systems
- **Reduce infrastructure costs** and simplify through tool consolidation
- **Accelerate cloud modernization** for greater flexibility and innovation
- **Elevate customer experiences**
- **Strengthen business process analytics** for data-driven decision making
- **Minimize cybersecurity risks** to protect assets and data

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Establishment of platform base for holistic AIOps capability scaling

To build a holistic AIOps capability,
smart monitoring across infrastructure, applications, and E2E chains
have to be combined.

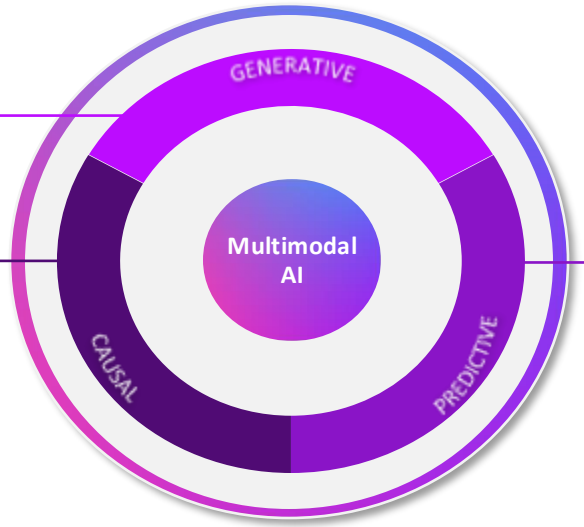


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Combining causal, predictive, and generative AI for modern AIOps

Multimodal AI

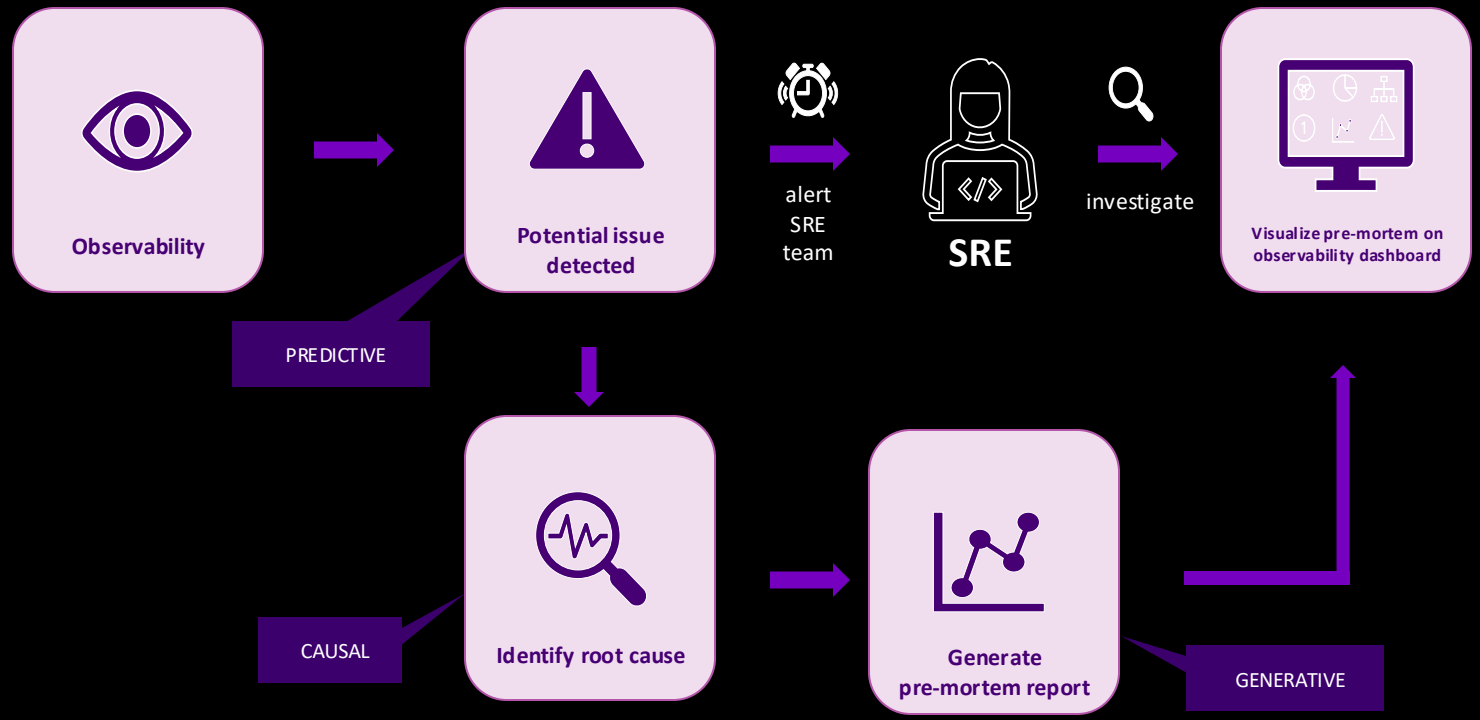
Produces responses, explanations, or remediation steps based on observed data and system context.



Identifies cause-effect relationships, enables RCA

Forecasts potential issues or performance trends based on historical patterns

Example: Proactively detecting potential problems in production and generating pre-mortem reports



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Demo: Zero-Touch AIOps

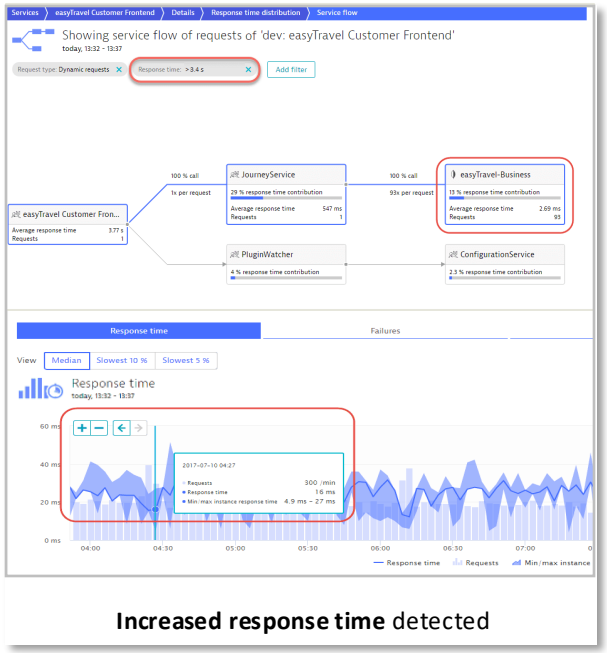
Shifting The Paradigm From Reactive To Predictive

Zero-Touch AIOps

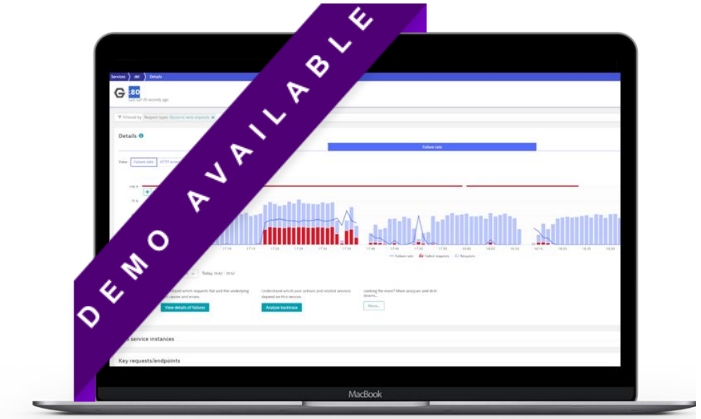
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Benefits

- Shift from reactive to predictive
- Minimize MTTR for common failures (self-healing)
- AI-powered root cause analysis
- Continuously measure reliability against SLOs, SLIs and other custom defined objectives
- Proactively alert teams before the failure occurs



Increased response time detected



AI

Problem Detected

Auto Remediation

Automatically scale server resources

Notify SRE team about problem resolution (e.g. update ticket status to 'Resolved')

SRE

Send **proactive alert to SRE team** (e.g. automatic ticket creation)

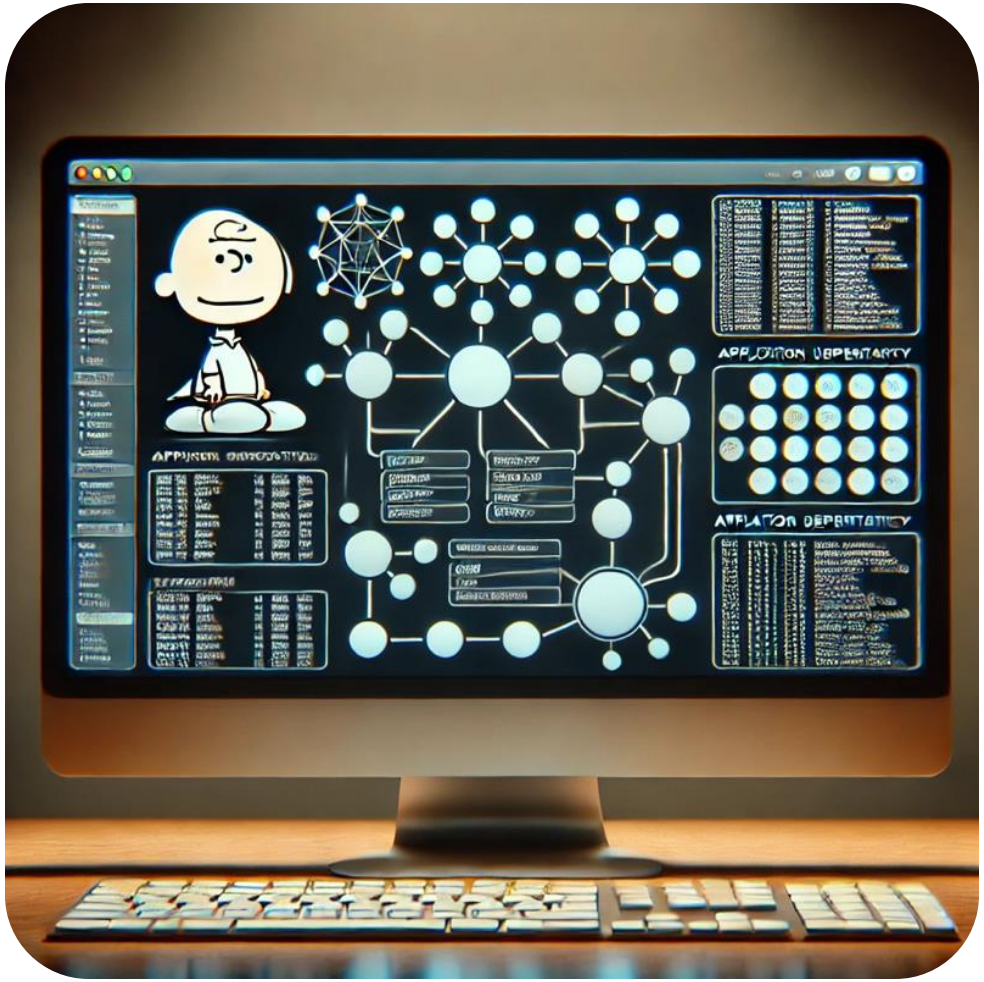
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Conclusion & Takeaways

Conclusion and Takeaways

- **AIOps** evolves from **contextualized observability** to AI-driven automation, enabling proactive insights, predictive analytics, and autonomous self-healing for optimized IT operations.
- Leveraging **multimodal AI** (causal, predictive, generative) can significantly elevate and strengthen your **Next-Gen AIOps** and **SRE** capabilities.
- Align IT operations with **business priorities** since the early stages of observability.
- Put your AIOps solutions to the test – **chaos experiments** can help you with that!



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Thank you!

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